

- 9 (a) product of density (of medium) and speed of sound (in medium) B1 [1]
- (b) difference in acoustic impedance M1
determines fraction of incident intensity
that is reflected/amount of reflection A1 [2]
- (c) pulse of ultrasound (directed into body) B1
reflected at boundary (between tissues) B1
(reflected pulse is) detected and processed B1
time for return of echo gives (information on) depth B1
amount of reflection gives information on tissue structures B1 [5]

- 10 (a) (i) amplitude (modulated) (allow AM) B1 [1]